

Theme 6: Vector Algebra

Note Title

2014/05/16

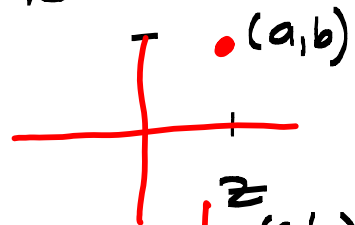
Unit 6.1 Three dimensional coordinate system. p 792

Notation $\mathbb{R}^n$

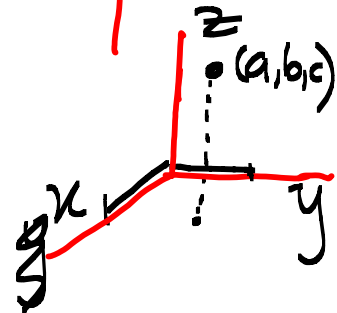
- One dimensional space ($\mathbb{R}^1$)
a line - with point a on the line



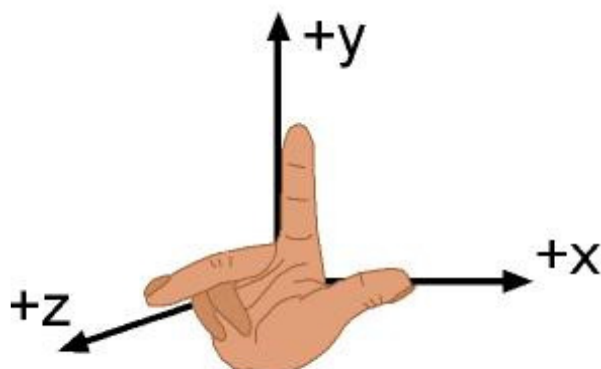
- Two dimensional space ($\mathbb{R}^2$)
a plane - with point (a,b)



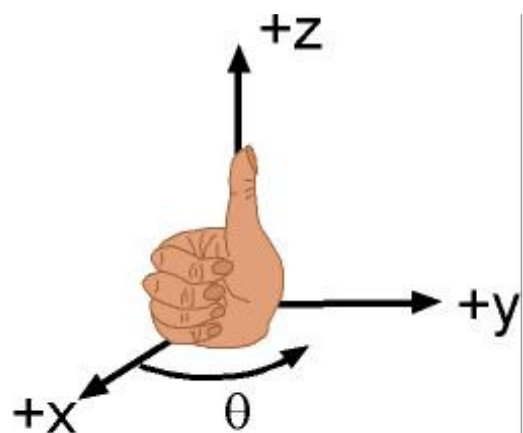
- Three dimensional space ($\mathbb{R}^3$)
space - with point (a,b,c)



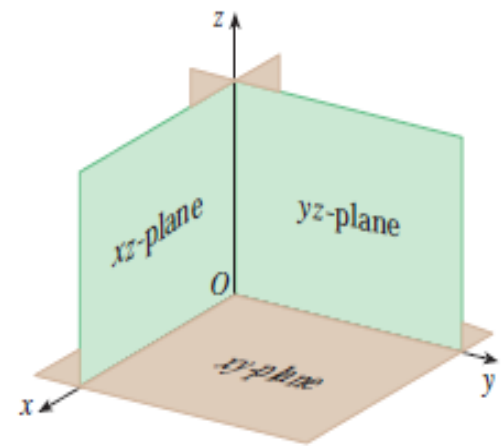
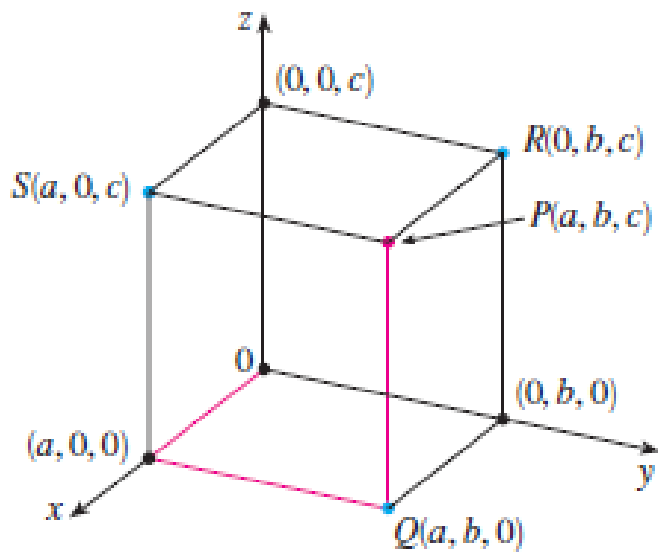
$$\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) / x, y, z \in \mathbb{R}\}$$



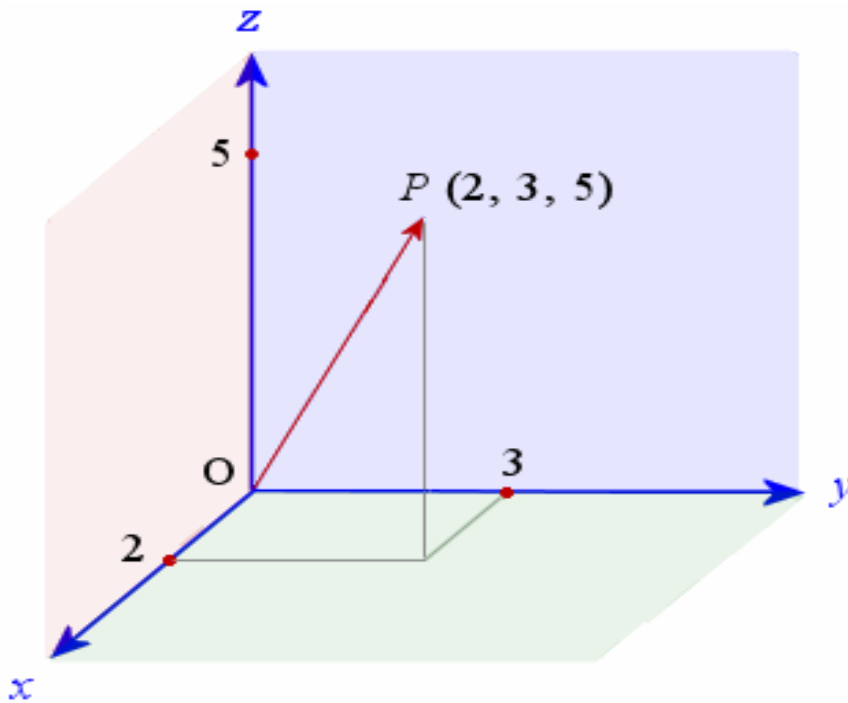
Cartesian
Right Hand Rule



Circular
Right Hand Rule

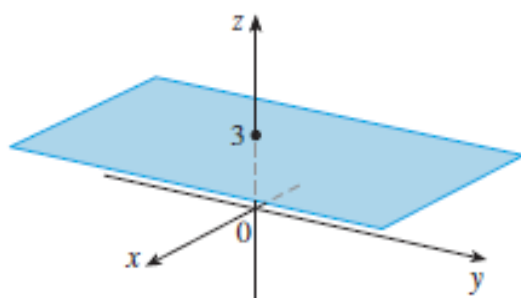


(a) Coordinate planes

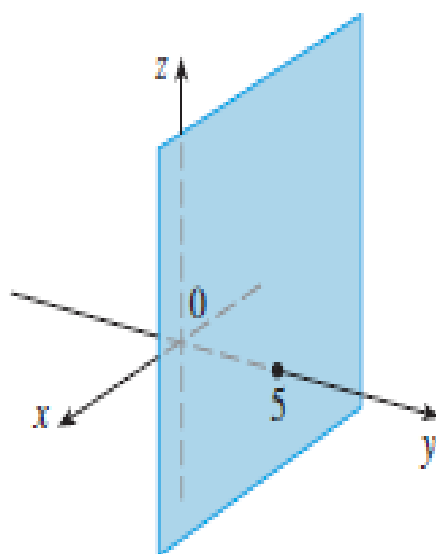


What surfaces in $\mathbb{R}^3$ are represented

- by :
- (a) $z = 3$
 - (b) $y = 5$
 - (c) $y = x$



1 (a) $z = 3$, a plane in $\mathbb{R}^3$



(b) $y = 5$, a plane in $\mathbb{R}^3$

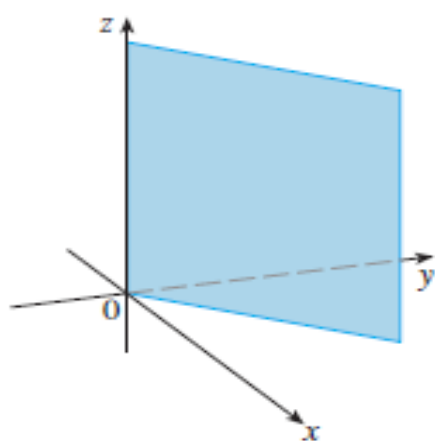
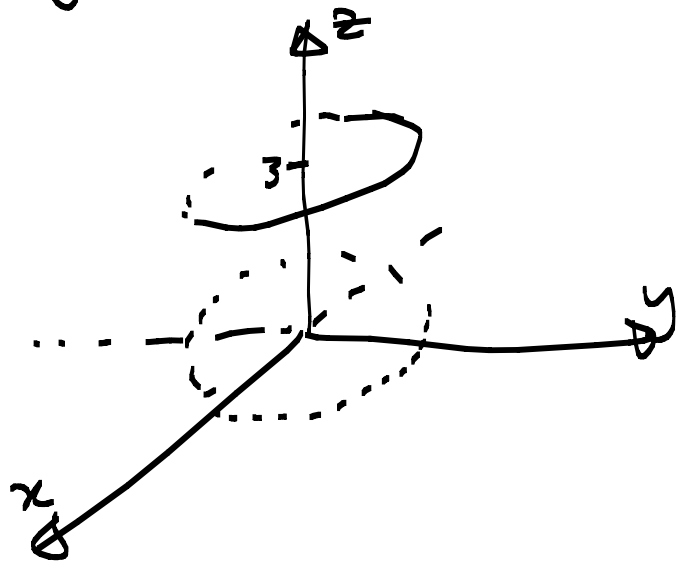


FIGURE 10
The plane $y = x$

What does $x^2 + y^2 = 1$ and $z = 3$ look like?



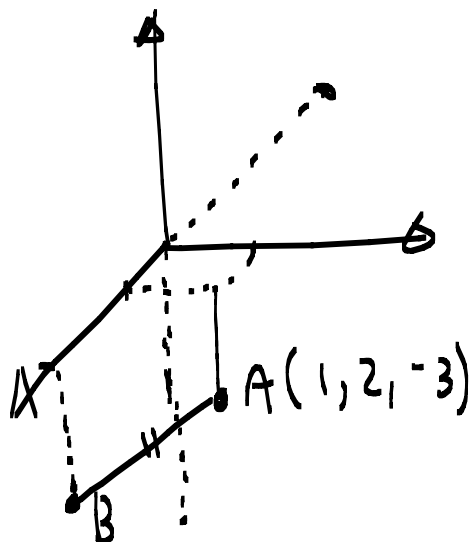
Distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ p795

$$|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

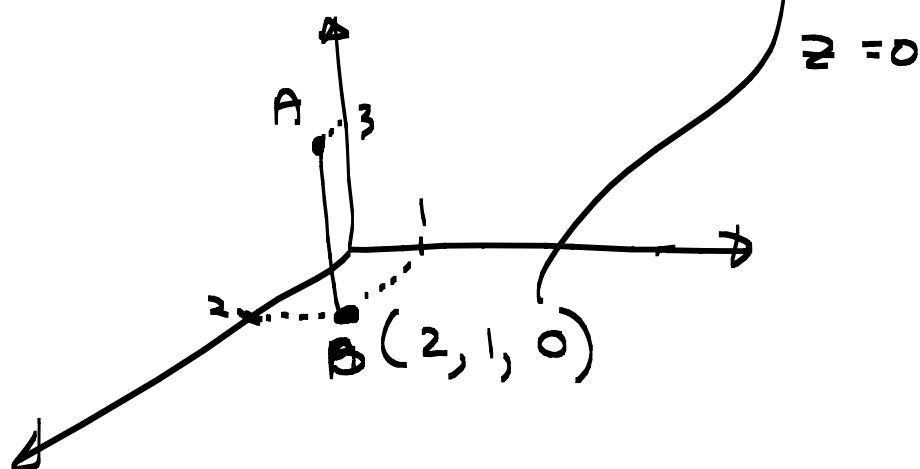
Example:

- ① Find the distance between $A(1, 2, -3)$ and $B(4, 0, -3)$

$$|AB| = \sqrt{(4-1)^2 + (0-2)^2 + (-3+3)^2} = \sqrt{13}.$$



② Distance $A(2, 1, 3)$ and xy -plane



$$|AB| = 3$$

Eq. of $\odot$; $(x-a)^2 + (y-b)^2 = r^2$
 (a, b) centre and r is radius.

Equation of a sphere; p 795

Radius r and center $C(h, k, l)$

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

Find the centre and radius of

the sphere:

$$x^2 + y^2 + z^2 + 2x - 6z + 8 = 0$$
$$x^2 + 2x + y^2 + z^2 - 6z = -8$$

Complete the square!

$$(x+1)^2 + y^2 + (z-3)^2 = -8 + 1 + 9$$
$$= 2$$

Sphere: radius $\sqrt{2}$ and centre $(-1, 0, 3)$